Expert-Guided Large Language Models for Clinical Decision Support in Precision Oncology

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ABSTRACT

PURPOSE Rapidly expanding medical literature challenges oncologists seeking targeted cancer therapies. General-purpose large language models (LLMs) lack domain-specific knowledge, limiting their clinical utility. This study introduces the LLM system Medical Evidence Retrieval and Data Integration for Tailored Healthcare (MEREDITH), designed to support treatment recommendations in precision oncology. Built on *Google's Gemini Pro* LLM, MEREDITH uses *retrieval-augmented generation* and *chain of thought*.

METHODS We evaluated MEREDITH on 10 publicly available fictional oncology cases with iterative feedback from a molecular tumor board (MTB) at a major German cancer center. Initially limited to *PubMed*-indexed literature (draft system), MEREDITH was enhanced to incorporate clinical studies on drug response within the specific tumor type, trial databases, drug approval status, and oncologic guidelines. The MTB provided a benchmark with manually curated treatment recommendations and assessed the clinical relevance of LLM-generated options (qualitative assessment). We measured semantic cosine similarity between LLM suggestions and clinician responses (quantitative assessment).

RESULTS MEREDITH identified a broader range of treatment options (median 4) compared with MTB experts (median 2). These options included therapies on the basis of preclinical data and combination treatments, expanding the treatment possibilities for consideration by the MTB. This broader approach was achieved by incorporating a curated medical data set that contextualized molecular targetability. Mirroring the approach MTB experts use to evaluate MTB cases improved the LLM's ability to generate relevant suggestions. This is supported by high concordance between LLM suggestions and expert recommendations (94.7% for the enhanced system) and a significant increase in semantic similarity from the draft to the enhanced system (from 0.71 to 0.76, $P = .01$).

CONCLUSION Expert feedback and domain-specific data augment LLM performance. Future research should investigate responsible LLM integration into real-world clinical workflows.

ACCOMPANYING CONTENT

Appendix

Data Sharing Statement

Visual Abstract

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INTRODUCTION

Somatic mutations are drivers of tumorigenesis.[1] Massively parallel sequencing has revolutionized our understanding of tumor biology.[2] However, translating this wealth of data into actionable clinical insights remains a challenge.[3] Molecular tumor boards (MTBs) play a crucial role in bridging this gap.[4,5] Personalized medicine approaches, enabled by molecular profiling, transform cancer treatment, particularly for advanced or rare tumors, by improving classification, extending progression-free survival, and potentially increasing overall survival rates.[6]

The challenge of translating genomic data into actionable treatment plans is amplified by the complex interplay of numerous, often parallel, genetic alterations without a clear driver mutation,[3] exacerbated by ever-expanding medical literature and fragmented data sources.[7,8] Current workflows require laborious curation of vast data sets, creating a bottleneck in precision oncology. Large language models

CONTEXT

Key Objective
Can domain-specific large language models (LLMs) support personalized treatment recommendations within molecular tumor boards (MTBs)?

Knowledge Generated
Using a comprehensive data set of research studies, oncologic guidelines, clinical trials, and drug availability, an LLM system was configured with retrieval-augmented generation and chain of thought to generate targeted treatment options for 10 fictional patients with cancer. The analysis showed high concordance between LLM-generated and expert oncologist recommendations, indicating that replicating the precision oncologists' evaluation process can enhance LLM performance for personalized cancer care.

Relevance
Domain-specific LLMs show potential for MTB decision support.

(LLMs) have the potential to streamline this process by extracting insights from diverse data sources, recognizing patterns, and aiding in personalized treatment plan development.[9]

In 2023, Benary et al[10] conducted a study to evaluate LLMs for developing personalized treatment plans for 10 fictional patients with advanced tumor diseases on the basis of their somatic mutation profiles. LLM-generated options were benchmarked with experts at a German cancer center. The study concluded that LLM-based treatment suggestions fell short of the quality and accuracy achieved by human experts.

This discrepancy was primarily attributable to LLMs trained on general knowledge and not medical literature,[11] neglecting contextual factors such as drug response in specific tumor types and drug availability.[12]

To address the limitations of general knowledge LLMs, we developed a *retrieval-augmented generation* (*RAG*) system[13,14] built on full-text articles retrieved from *PubMed* through open or institutional access. Our system Medical Evidence Retrieval and Data Integration for Tailored Healthcare (MEREDITH) was further enhanced by incorporating several data sources, mirroring the approach MTB experts use to evaluate and recommend personalized treatment plans: *PubMed*-indexed clinical studies on drug response within the specific tumor type, trial databases, authorized medication lists for drug availability, and oncologic guidelines. This enabled a general purpose LLM (*Google's Gemini Pro*) to evaluate and generate evidence-based molecularly targeted treatment options in an approach similar to MTB experts.

The MTB at the Center for Personalized Medicine at Technical University of Munich (ZPM^{TUM}) evaluated LLM-generated treatment options. This study investigates the potential of LLMs to support clinical decision making by MTBs.

METHODS

To evaluate the degree to which LLM-generated treatment suggestions complement the recommendations of medical experts, we used 10 publicly available fictional oncologic case vignettes.[10] The study adhered to the *Standards for Reporting of Diagnostic Accuracy* guidelines. Ethical clearance was not mandated as no patient data were used.

A comprehensive outline of the study workflow is presented in Appendix Figure A1.

Description of Fictional Case Vignettes

Benary et al[10] developed a collection of 10 fictional oncologic case vignettes, four of which were previously published to investigate the variability in molecularly targeted therapies among different raters.[15] The vignettes depicted seven tumor types and 59 molecular alterations. Lung adenocarcinoma constituted a large portion of the collection because of the widespread use of multigene sequencing.

The 10 patients are fully documented in Appendix Table A1.

Data Processing

The American College of Medical Genetics and Genomics and the Association for Molecular Pathology established a five-tier classification for the pathogenicity of genetic variants.[16] Benary et al[10] did not explicitly categorize variant pathogenicity.

Appendix Figure A2 illustrates the operational workflow of the MTB at ZPM^{TUM}.

ZPM^{TUM} pathologists generate standardized reports encompassing tumor profiling and variant pathogenicity.[8]

All 59 alterations received a pathogenicity class (1-5) before manual clinical annotation by MTB experts.

LLM System Architecture

Using LLMs requires prompts that describe the task in natural language. The field of generative artificial intelligence (AI) research is rapidly evolving, with commercially available and open-source models released at fast pace. However, meta studies suggest that the chosen LLM may be less critical than the configuration and evaluation mechanisms used to validate results.[17] As a previous study found *Google*'s *gemini-1.0-pro-001* to perform as well as humans in expert-guided data extraction and summarization of complex, domain-specific language,[18] this LLM was selected for the study.[19]

LLMs are trained on data sets reaching sizes of hundreds of terabytes. This training process imbues them with statistical representations of language patterns, which enable LLMs to perform tasks on the basis of natural language prompts. However, question-answering systems built with LLMs can exhibit limitations, including factually incorrect responses (hallucinations) and the inability to cite sources (lack of attribution). Two approaches have emerged to address these challenges: fine-tuning for improved closed-domain question answering and RAG for incorporating additional memory.[13] *RAG* consists of two components: a generator that enhances LLMs by adding relevant factual information (context) to the input, and a retriever that selects context on the basis of its relevance to the query. We compiled a corpus of scientific literature from *PubMed* using the *PyMed* application programming interface with a targeted search string specific to patient diagnosis and tumor profile (eg, non–small cell lung cancer + *KRAS* G13D + targeted therapy).[20]

After expert feedback to contextualize molecular targets, the document corpus was enriched with clinical data sources, including *PubMed*-indexed clinical studies on drug response within the specific tumor type, clinical trials obtained from *ClinicalTrials.gov* and *QuickQueck.de*, guidelines from oncology associations (*ASCO*, *European Society for Medical Oncology*, and *National Comprehensive Cancer Network*), German national guidelines (*Deutsche Krebsgesellschaft*, *Deutsche Gesellschaft für Hämatologie und Medizinische Onkologie*), and authorized medication lists (*European Medicines Agency* [*EMA*], and *US Food and Drug Administration* [*FDA*]). Retrieval strategies prioritized recent publications and options with robust supporting evidence, as defined by the established National Center for Tumor Diseases and German Consortium for Cancer Research framework (m1A indicating strongest evidence).[21] This approach ensured researchers received up to five relevant documents along with evidence levels, facilitating informed decision making.

To enhance model precision, *chain of thought* (*CoT*) was used. *CoT* enables reasoning in complex decision-making scenarios.[22] Four prompts guided the model to summarize literature, identify applicable guidelines and drug availability, identify recruiting trials, and synthesize findings into recommendations.

Figure 1 illustrates system architecture for MEREDITH.

Clinical Annotation of the Case Vignettes

Clinical annotation evaluates oncology guidelines, published research on molecular alterations within the same or different tumor types, functional implications, roles in the overall biological pathway, rationales for molecularly targeted therapy, and availability of corresponding drugs through *EMA*/*FDA* approval or locally available clinical trials.

The case vignettes underwent a two-stage evaluation by the MTB team, comprising biologists, clinicians, geneticists, and pathologists. The MTB panel presented and discussed their own recommendations before evaluating LLM-generated options.

Qualitative and Quantitative Assessment of the LLM-Generated Recommendations

A multidisciplinary team reviewed patients one through five in one session and patients six through ten in a subsequent session, separated by 1 week. This two-part review allowed for qualitative evaluation of LLM-generated therapy options and model refinement during the intervening week. To mitigate confirmation bias, LLM developers received benchmark expert responses during final case discussions. The first session presented a draft model on the basis of *PubMed*-indexed literature. Leveraging ZPM^{TUM} expertise, the enhanced version analyzes a wider range of knowledge sources, enabling data-driven validation of patient-specific treatment options considering guidelines, drugs, and trials. To ensure unbiased evaluation by MTB members, they were not informed of LLM system enhancements on the basis of their feedback from the first session.

Patient-level data on (likely) pathogenic molecular alterations and recommended treatment options were summarized using median and IQR.

The MTB panel assessed LLM-generated treatment options for alignment with guidelines and trials, consistency with human specialist recommendations, and potential biases or factual inconsistencies introduced by the LLM system.

To quantitatively assess agreement between LLM-generated options and clinician responses, we used cosine similarity. This technique compares numerical representations of sentences, known as text embeddings, which represent meaning. Cosine similarity for text embeddings is in a range of $0.0 < \cos(\theta) \leq 1.0$. Words that change the meaning of a sentence, such as not, reduce cosine similarity. Higher cosine similarity scores indicate closer semantic similarity between

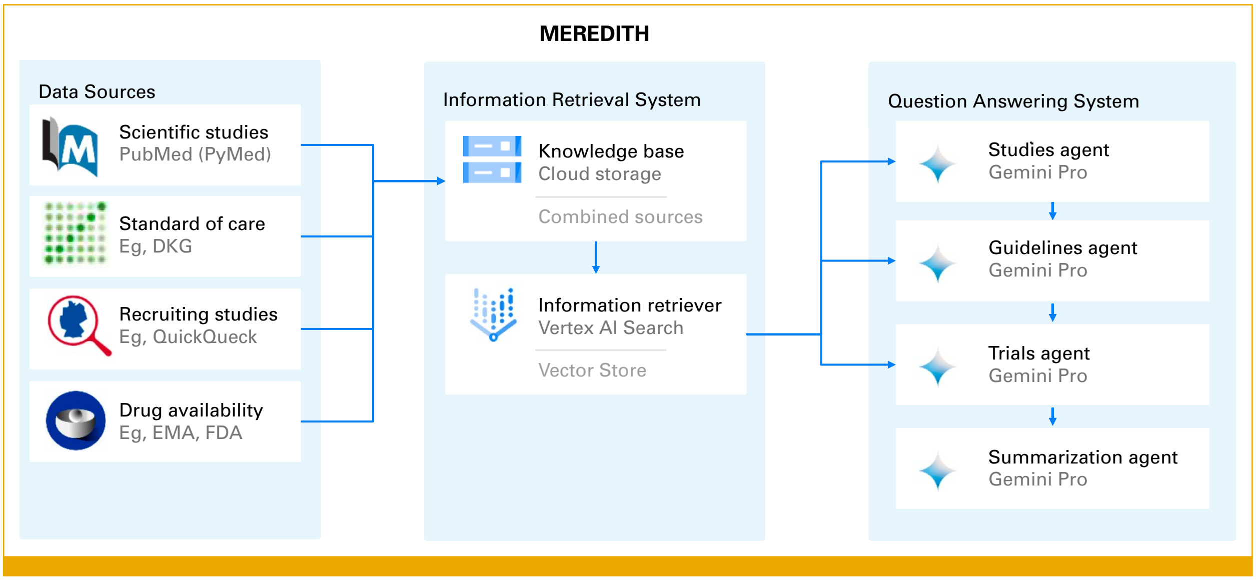

FIG 1. Architecture of the enhanced LLM system MEREDITH. DKG, Deutsche Krebsgesellschaft (German Cancer Society); EMA, European Medicines Agency; FDA, US Food and Drug Administration; LLM, large language model; MEREDITH, Medical Evidence Retrieval and Data Integration for Tailored Healthcare.

LLM suggestions and clinician recommendations. To evaluate the impact of model enhancements, treatment suggestions for all 10 patients were generated using both draft model and enhanced model. Cosine similarity was calculated to compare the outputs between LLM suggestions and clinician recommendations. We then used a paired *t*-test to evaluate the statistical significance of improvements in similarity scores. All analyses were conducted in *Python*, with a significance level of $\alpha = .05$.

All scripts and raw data are available in the study GitHub repository.[23]

RESULTS

LLM Options Were Semantically Similar to Expert Recommendations

Ten hypothetical patients with cancer were evaluated for molecularly targeted therapies (Appendix Table A1). Median age was 57 years (range, 44.75-60.25). Gender data were available for 60% (n = 6), with an equal distribution of males and females. Patients harbored a median of 3.5 molecular alterations (range, 3.0-4.8) per patient, with a total of 59 identified across all patients. Of these alterations, the median number classified as (likely) pathogenic was 2.5 per patient (range, 1.75-4.25). Given the limited clinical data in the case vignettes, treatment recommendations were generated under the assumption that standard therapies had been administered.

Expert review identified a molecular rationale for 18 of 33 (likely) pathogenic alterations, while the LLM systems identified a rationale for 21.

Table 1 summarizes the number of proposed treatment options, molecular rationales for treatment response/resistance, and potential molecular interactions for each patient, as identified by medical experts and the LLM systems (draft and enhanced).

The median number of treatment options proposed per patient was 2.0 for MTB experts (IQR, 1.0-3.0), 3.5 for the draft LLM system (IQR, 2.0-4.0), and 4.0 for the enhanced LLM system (IQR, 3.0-5.0). Both LLM systems proposed a broader range of treatment options compared with experts.

TABLE 1. Number of Proposed Treatment Options, Molecular Rationales for Treatment Response/Resistance, and Potential Molecular Interactions for Each, as Identified by Medical Experts and the LLM Systems (draft + enhanced)

Patient	Medical Experts	Draft System[a]	Enhanced System[b]
1	1	2	2
2	1	4	3
3	4	5	6
4	2	2	4
5	2	3	5
6	2	2	3
7	1	4	4
8	2	2	3
9	3	4	5
10	3	4	4

Abbreviation: LLM, large language model.
[a]The draft LLM system was discussed by medical experts for patients 1-5 only.
[b]The enhanced LLM system was discussed by medical experts for patients 6-10 only.

This difference was particularly evident in recommendations on the basis of preclinical evidence and combination therapies.

For patients $x = [1, 10]$, the mean cosine similarity $m(\cos(\theta))$ between MTB expert recommendation and iteration one of the LLM system was $m(\cos(\theta)) = 0.71$, IQR [0.61-0.80]. For patients $x = [1, 10]$, the mean cosine similarity $m(\cos(\theta))$ between MTB expert recommendation and iteration two of the LLM system was $m(\cos(\theta)) = 0.76$, IQR [0.73-0.82]. That means cosine similarity increased by an average of 9% from iteration 1 to iteration 2. The paired t-test showed this increase to be significant with $P = .01$. See Table 2 for detailed results.

Comprehensive Data Improve LLM Alignment With Expert Recommendations

Table 3 summarizes the qualitative assessment results as reviewed by the MTB panel for each patient, categorized by the LLM system (draft or enhanced) that generated the treatment suggestion.

Figure 2 provides a visual representation of the qualitative assessment results for LLM-generated treatment options, as detailed in Table 3. It categorizes the results by the LLM system used (draft or enhanced).

Overall, the enhanced LLM system showed improved concordance with MTB panel recommendations because of the integration of clinical knowledge and drug availability data. This enabled the LLM to replicate the contextualization of molecular targets achieved by experts.

Agreement

The LLM system achieved high concordance with the MTB panel (94.7% for enhanced model; 77.1% for both models). This included targeted therapies with evidence from studies within the same cancer type (m1A, m1B) or from large-scale studies in different cancer types (m2A) and immunotherapy for patients with high tumor mutational burden. To explore a wider range of treatment options, the model was configured to generate diverse options (temperature = 0.2, Top-P = 1.0, Top-K = 32). Expert review and prioritization ensured patient suitability. The system identified options on the basis of preclinical data (m3) and combination therapies, which were validated by the MTB panel for clinical relevance after being absent from expert recommendations. This suggests

TABLE 2. Quantitative Assessment of the Level of Agreement Between Large Language Model Recommendations and Clinician Responses Using Cosine Similarity

Patient	Target	cos (θ) Draft System[a]	cos (θ) Enhanced System	Δ
1	*KRAS* p.G13D	0.61	0.77	0.16
1	*TP53* p.A276G	0.63	0.63	0.00
2	*PIK3CA* amplification (n > 6)	0.62	0.61	0.01
2	*MYC* amplification (n > 6)	0.61	0.75	0.14
2	*SOX2* amplification (n > 6)	0.54	0.75	0.22
3	*ATM* p.E1666* (4N)	0.58	0.82	0.23
3	*ERBB2* amplification	0.84	0.84	0.00
3	*FGFR3* amplification	0.75	0.73	0.02
3	*MET* amplification	0.80	0.78	0.02
4	*PIK3CA* p.E545K mutation	0.78	0.77	0.01
4	*MAPK1* p.E322K mutation	0.61	0.62	0.01
5	*EGFR* mutation	0.84	0.78	0.06
5	*MET* amplification	0.75	0.83	0.08
6	*KRAS* p.G12C mutation (AF 18%)	0.79	0.82	0.03
7	*NF1* p.I1605fs mutation (AF 39%)	0.84	0.80	0.04
7	*TMB* 12.8 Mut/Mb	0.62	0.64	0.02
8	*FGFR2::BICC1* fusion	0.73	0.76	0.03
9	*TMB* 10.5 Mut/Mb	0.75	0.76	0.01
9	*HRAS* p.Q61R mutation *PIK3CA* comutation	0.73	0.84	0.11
9	*PIK3CA* p.E545K (AF 39%)	0.82	0.84	0.02
10	*EGFR* p.E746_A750del mutation (AF 50%)	0.60	0.63	0.03
10	*EGFR* p.C797S mutation	0.74	0.73	0.01
10	*STK11* p.C210* mutation (AF 39%)	0.80	0.93	0.13

Abbreviation: AF, allele frequency.
[a]Only patients 1-5 were discussed during iteration 1.

TABLE 3. Qualitative Assessment of the LLM-Generated Treatment Suggestions as Reviewed by the MTB Experts

Case	MTB Panel Review of the LLM-Generated Treatment Suggestions
1	A 56-year-old patient with lung adenocarcinoma (Likely) pathogenic molecular alterations: ***KRAS* p.G13D**, ***TP53* p.A276G** Draft LLM system: *KRAS* G13D mutation is a potential target for cetuximab therapy in colon cancer (m2B). Anti–PD-1/PD-L1 therapy: *KRAS* and *TP53* missense comutations are associated with a better response to anti–PD-1/PD-L1 therapy MTB panel review: disagreement: cetuximab is not recommended because of lack of supporting phase III data in patients with NSCLC
2	A 26-year-old patient with urachal carcinoma (Likely) pathogenic molecular alterations: ***KRAS* p.G12V**, ***TP53* p.H214fs*7**, ***MYCN* p.E47fs*8**, ***PIK3CA* amplification (n > 6)**, ***MYC* amplification (n > 6)**, ***MYCL1* amplification (n > 6)**, ***SOX2* amplification (n > 6)**, ***MUTYH* amplification (n>6)** Draft LLM system: *PIK3CA* amplification is a potential target for everolimus or alpelisib therapy. *MYC* amplifications are not observed in *TP53*-WT patients. Presence of *MYC* amplification in the tumor cells confers refractoriness to treatments with anti–PD-1 and anti–PD-L1 (m2R). *MYC* amplification is a potential target for aurora kinase inhibition (m2C, m3) and *CHK1* inhibition (m2C). *SOX2* amplification leads to increased *CDK7* protein levels. Inhibition of *CDK7* can lead to decreased SOX2 target gene expression and decreased cell proliferation (m3) MTB panel review: concordance: everolimus may be considered following the exhaustion of available treatment options. Disagreement: aurora kinase inhibitors, CHK1 inhibitors, and CDK7 inhibitors are currently unavailable
3	A 58-year-old patient with thymic carcinoma (Likely) pathogenic molecular alterations: ***SMAD4* p.C363R (1N)**, ***ATM* p.E1666* (4N)**, ***PRKDC-CDH17* fusion**, ***EXT1-MAGI2* fusion**, ***ERBB2* RNA overexpression (RPKM tumor 45 *v* 1.8 control)**, ***ERBB3* RNA overexpression (RPKM tumor 65.9 *v* 0.2 control)**, ***PDGFRB* RNA overexpression (RPKM tumor 35.8 *v* 2.3 control)**, ***TGFA* RNA overexpression (RPKM tumor 14.2 *v* 0.4 control**, ***EGF* RNA overexpression (RPKM tumor 1.9 *v* 0.1 control)**, ***FGFR3* RNA overexpression (RPKM tumor 11.4 *v* 1.9 control)**, ***MET* RNA overexpression (RPKM tumor 22.1 *v* 1.4 control)** Draft LLM system: *ERBB2* amplification is a potential target for trastuzumab deruxtecan, trastuzumab, pertuzumab, ado-trastuzumab emtansine (T-DM1) (m2A). *FGFR3* overexpression is a potential target for AZD4547, infigratinib (pan-FGFR inhibitor), and futibatinib (FGFR1-4 inhibitor) therapy (m2A). Increased expression of p*ERBB3* is a key mechanism of adaptive resistance to *FGFR* inhibitors. Resistance can be overcome by combination treatment with a pan-ERBB inhibitor (m3R). *MET* amplification is associated with increased benefit from *MET*-directed targeted therapy, eg, capmatinib. The combination of *MET* inhibitor together with ICB may overcome ICB resistance induced by *MET* amplification (combinatory treatment) MTB panel review: concordance on *MET*, *HER2*, and *FGFR3* overexpression; similar treatment recommendations provided Disagreement: infigratinib + AZD4547 are currently unavailable. The LLM system did not make a recommendation regarding HRD testing or the use of PARP inhibitors in the context of the *ATM* mutation
4	A 59-year-old patient with oropharyngeal carcinoma (Likely) pathogenic molecular alterations: ***PIK3CA* p.E545K (AF 25%)**, ***MAPK1* p.E322K (AF 10%)** Draft LLM system: *PIK3CA* hotspot mutation is a potential target for everolimus or alpelisib therapy (m2A). *MAPK1* mutation: in a window-of-opportunity clinical trial, a patient with a stage IVA tongue carcinoma who received a 13-day course of erlotinib as first-line treatment experienced remarkable disease reduction (m2C, m3) MTB panel review: concordance on *PIK3CA* mutation; similar treatment recommendation provided Neutral: previous therapies are critical for determining the suitability of erlotinib treatment (TKIs are approved for the treatment of oropharyngeal carcinoma by both the EMA and the FDA)
5	A 64-year-old female patient with lung adenocarcinoma (Likely) pathogenic molecular alterations: ***EGFR* p.E746_A750del (AF 43%)**, ***MET* amplification FISH positive** Draft LLM system: although *EGFR* inhibitors have revolutionized treatment for *EGFR*-mutated lung cancers, resistance often emerges because of *MET* amplification. Combining TKI therapy and *EGFR-MET* bispecific antibodies holds promise for overcoming resistance and extending patient outcomes (m1B). *MET* amplification is associated with increased benefit from *MET*-directed targeted therapy, eg, capmatinib. The combination of *MET* inhibitor together with ICB may overcome ICB resistance induced by *MET* amplification (combinatory treatment) MTB panel review: concordance on *MET* amplification; similar treatment recommendation provided
6	A 59-year-old male patient with lung adenocarcinoma (Likely) pathogenic molecular alterations: ***KRAS* p.G12C (AF 18%)** Enhanced LLM system: sotorasib is a *KRAS G12C* inhibitor that has been shown to be effective in treating patients with *KRAS G12C*-mutated lung cancer. Currently recruiting studies in Germany: 1. KRYSTAL-12, NCT04685135, 2. KontRASt-01, NCT04699188, 3. NCT05789082, 4. KontRASt-03, NCT05358249. Comutation of *STK11* and *KEAP1* was associated with significantly shorter survival, independent of KRAS G12C mutational status (m1B) MTB panel review: concordance with the MTB panel on treatment recommendations
7	A 41-year-old female patient with melanoma (Likely) pathogenic molecular alterations: **tumor mutational burden 12.8 Mut/Mb**, ***NF1* p.I1605fs (AF 39%)**, ***TP53* c.672+1G>A (AF 50%)** Enhanced LLM system: in the presence of an *NF1* mutation, selumetinib or cabozantinib monotherapy (m2B) or combination therapy with dacarbazine (1Amu) may be considered. TMB ≥10 Mut/Mb suggests potential benefit from immunotherapy MTB panel review: treatment recommendations largely aligned with the MTB panel. Recommendations diverged regarding the MEK inhibitor. The MTB panel prefers trametinib (over selumetinib) for monotherapy The LLM-proposed combination therapies warrant further investigation and may be prioritized on the basis of previous treatment history
8	A 46-year-old male patient with cholangiocarcinoma (Likely) pathogenic molecular alterations: ***FGFR2::BICC1* Fusion**, ***TP53* p.E258* (AF 52%)** Enhanced LLM system: for FGFR2 fusion-positive cholangiocarcinoma, futibatinib and pemigatinib are currently approved treatment options (m1A). Infigratinib, although not yet approved by the EMA, is under expedited review and holds potential for future approval. **Currently recruiting study in Germany:** REFOCUS, NCT04526106 MTB panel review: concordance with the MTB panel on treatment recommendations
9	A 79-year-old male patient with salivary duct carcinoma (Likely) pathogenic molecular alterations: **tumor mutational burden 10.5 Mut/Mb**, ***HRAS* p.Q61R (AF 44%)**, ***PIK3CA* p.E545K (AF 39%)** Enhanced LLM system: TMB ≥10 Mut/Mb suggests potential benefit from immunotherapy. **Currently recruiting study in Germany:** MK-3475-158/KEYNOTE-158, NCT02628067, CRAFT, Arm 7, NCT04551521 (the patient is not eligible for this study arm because of the presence of a *PIK3CA* hotspot mutation, consider enrollment in Arm 5). The farnesyltransferase inhibitor tipifarnib as a single agent was administered in 6 patients with

(continued on following page)

TABLE 3. Qualitative Assessment of the LLM-Generated Treatment Suggestions as Reviewed by the MTB Experts (continued)

Case	MTB Panel Review of the LLM-Generated Treatment Suggestions
	HNSCC. Among 5 patients with available data, 1 PR, 2 SD, and 2 PD were achieved as best responses. PFS data were available for 6 patients and PFS was more than 6 months in 3 patients (m1C). Tipifarnib is not currently available for salivary duct carcinoma in Europe. *PIK3CA* mutations may confer a higher risk of developing resistance to *HRAS*-targeted therapies, such as MEK inhibitors. Tipifarnib potentiates the antitumor effects of PI3Kα inhibition in *PIK3CA*- and *HRAS*-dysregulated HNSCC via convergent inhibition of mTOR activity (m2B). *PIK3CA* hotspot mutation is a potential target for everolimus, temsirolimus, or alpelisib therapy. Currently recruiting studies in Germany: CRAFT, Arm 5, NCT04551521, TAPISTRY, Arm H, NCT04589845 MTB panel review: concordance with the MTB panel on treatment recommendations The LLM system accurately highlighted the current unavailability of tipifarnib
10	A 52-year-old female patient with lung adenocarcinoma (Likely) pathogenic molecular alterations: ***EGFR* p.E746_A750del (AF 50%)**, ***EGFR* p.C797S (AF 29%)**, ***STK11* p.C210* (AF 39%)** Enhanced LLM system: the combination of amivantamab and lazertinib has been shown to be effective in patients with *EGFR* exon 19 deletion mutations, including the p.E746_A750del mutation (m1A). Amivantamab is an EGFR-MET bispecific antibody that has shown promising activity in patients with *EGFR*-mutant NSCLC who have progressed on previous EGFR TKIs. Lazertinib is a third-generation *EGFR* TKI that has been shown to be effective in patients with *EGFR*-mutant NSCLC who have progressed on previous *EGFR* TKIs. The combination of amivantamab and lazertinib has been shown to be effective in patients with *EGFR*-mutant NSCLC who have progressed on previous *EGFR* TKIs, including those with the *EGFR* p.C797S mutation (m1A). As of today, lazertinib and amivantamab have not yet been approved by the EMA. Although still undergoing review by the EMA, patients might consider enrolling in a clinical trial (NCT04077463) currently recruiting participants in Germany to explore the potential benefits of these new drugs. **Currently recruiting study in Germany:** Chrysalis-2, NCT04077463. *STK11* mutations are associated with worse responses to immunotherapy MTB panel review: concordance with the MTB panel on treatment recommendations The LLM system additionally identified the availability of the amivantamab + lazertinib combination within a clinical trial, offering a potential treatment avenue

NOTE. The entries highlighted in bold are (likely) pathogenic genetic alterations.
Abbreviations: AF, allele frequency; EMA, European Medicines Agency; FDA, US Food and Drug Administration; HNSCC, head and neck squamous cell carcinoma; LLM, large language model; MTB, molecular tumor board; NSCLC, non–small cell lung cancer; TKI, tyrosine kinase inhibitor.

potential for domain-specific LLM systems to support evidence-based treatment decisions.

Disagreement

The MTB panel disagreed with none (0%) of the enhanced LLM-generated treatment options. All six disagreements (17.1%) originated from the draft model. The MTB panel disapproved use of cetuximab because of insufficient data in certain patient contexts. The draft LLM proposed investigational drugs (eg, aurora kinase inhibitors) that are not yet clinically available, highlighting the importance of considering practical limitations. The draft LLM system did not recommend HRD testing for *ATM* mutations, a potential indicator for PARP inhibitor therapy. This underlines the need for continuous improvement in LLM abilities to address relevant clinical factors.

Neutral Stance

The MTB panel neither agreed nor disagreed on two LLM-proposed treatments (5.7%): erlotinib for a *MAPK1* mutation and tyrosine kinase inhibitors (TKIs) for an *NF1* mutation that deviated from the expert-recommended TKI. This

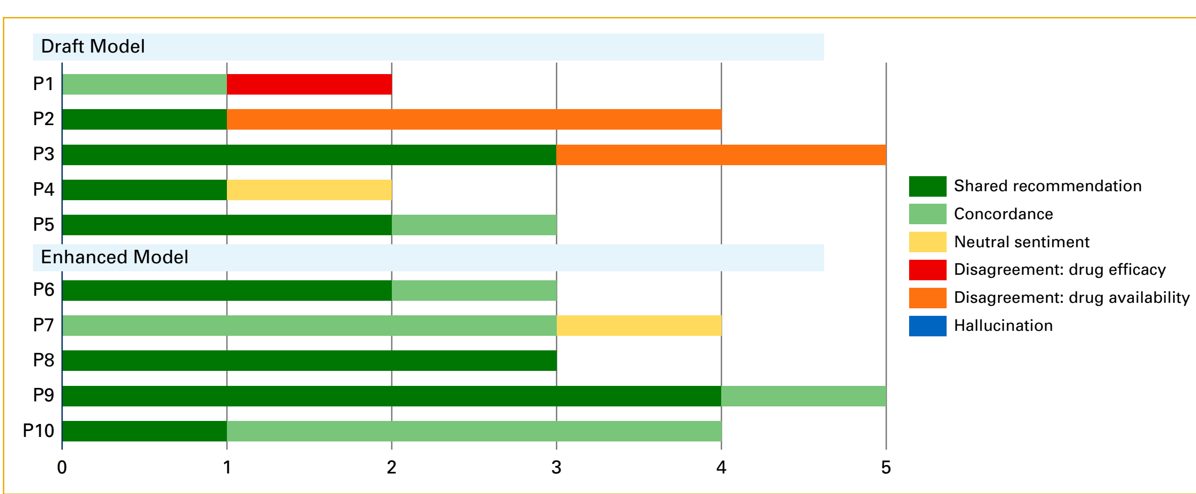

FIG 2. Qualitative assessment of LLM-generated treatment options by expert panel. LLM, large language model.

underscores the importance of incorporating previous treatment history and clinical expertise in personalized therapy selection.

Additionally, the enhanced LLM system provided information on drug availability and clinical trials, offering context and alternative options for patients. The LLM system's ability to explain the interplay between different molecular targets offers a scientific rationale for treatment response and resistance. The MTB panel suggested additional diagnostic tests not included in this analysis, such as tumor or germline analysis (detailed in Appendix 1).

Expert-Guided System Design Mitigates Hallucination Risk

By following expert guidance, the LLM system MEREDITH accurately referenced literature without presenting factually incorrect information (hallucinations) in all cases. The enhanced LLM system successfully retrieved all references identified by human experts, suggesting potential to streamline literature reviews for researchers.

DISCUSSION

This study investigated the potential of the LLM system MEREDITH to support personalized treatment recommendations in precision oncology. Agreement between LLM-generated options and MTB experts was evaluated using 10 fictional cancer case studies with 59 molecular alterations. Analysis revealed significant improvement in qualitative agreement with MTB experts and quantitative performance metrics compared with the draft LLM system. The enhanced LLM facilitated exploration of treatment options, especially for preclinical studies and treatment combinations subsequently validated by the MTB panel. It consistently generated correctly referenced options. This success underscores the value of incorporating data sources such as clinical response studies, oncology guidelines, and trial databases in LLM refinement that was guided by MTB experts.

MEREDITH integrates expert reasoning with domain-specific data, aiming to improve LLM accuracy for precision oncology applications.[24] The enhanced LLM achieved substantial agreement with expert oncologists with 52.6% exact matches of treatment recommendations and an expert concordance rate of 94.7% when evaluating LLM-generated suggestions not found in the benchmark. This performance is comparable with real-world interrater agreement of 66% among MTBs.[25] Even experienced experts reach diverse conclusions because of the complexity of oncology treatment decisions.[15] Our study and real-world MTBs observed deviations primarily in preclinical data and combination therapies. This suggests areas where further research and standardization efforts may be particularly valuable.

This study replicated a real-world MTB workflow with expert assessment and iterative refinement. A previous study compared the performance of four general-purpose LLMs for treatment recommendations in precision oncology.[10] A key limitation was the lack of broader context in recommendations, as MTB experts highlighted the need for options that consider not only the molecular alteration but also factors such as drug response within the same tumor type and drug availability. Our LLM system MEREDITH addressed this limitation by generating treatment options that were relevant for the tumor type, clinically actionable with available treatments (in-line or off-label use), options within ongoing clinical trials, and literature for evaluation by human experts. Our rigorous development process, including blinded evaluations and delayed access to benchmark expert responses, limited bias on the LLM system. This enabled assessing MEREDITH's ability to generate treatment options autonomously. To ensure the best possible patient care, we leveraged a human-in-the-loop approach. Our LLM system offered a wide range of treatment options, which were reviewed and prioritized by experts for specific patient needs. MEREDITH's capability to generate evidence-based argumentation for off-label therapies has the potential to expedite the creation of comprehensive cost-coverage justification reports. This could streamline communication with insurers, facilitating faster patient access to clinically relevant treatment options.

Although MEREDITH's information corpus was comprehensive, a significant portion required proprietary institutional access. This reliance on non–open-access resources could hinder broader adoption of our LLM system. The development of open-access repositories for domain-specific scientific literature is crucial for overcoming limitations of LLM systems built on paywalled data.[26] The use of hypothetical scenarios is a limitation.[27] After resolution of privacy and confidentiality concerns, we plan to assess the LLM system's performance in a real-world setting using anonymized patient data.

The application of LLMs in precision oncology raises ethical concerns centered on patient privacy.[28,29] Robust anonymization techniques and secure data storage are essential to mitigate risks associated with patient data in LLMs. Balancing innovation in oncologic care and safeguarding patient well-being is paramount. This necessitates ongoing monitoring of LLM development and deployment, responsible data handling practices to ensure patient data protection, and explainable AI (XAI) to promote transparency and trust in LLM decision-making processes.[30,31] LLM-provided literature references enhance health care professionals' understanding of treatment rationale, promoting trust in generated options.

Integration of LLMs in MTBs offers several advantages. LLM systems enhanced with relevant data provide access to oncologic research, facilitating evidence-based decision making and serving as a useful assistant for physicians. Advanced data analysis alleviates information overload for clinicians, allowing for focused attention on individual cases.

LLMs have the potential to democratize access to expert advice via telemedicine and provide knowledge resources for complex cases. LLMs such as MEREDITH can further enhance MTB capabilities by promoting continuous knowledge acquisition within the rapidly evolving field of oncology. They may also contribute to data standardization, potentially improving training data quality.[6] LLMs hold promise for augmenting MTB decision-making capabilities, while human experts retain ultimate responsibility for treatment selection.

In conclusion, this study demonstrates the potential of domain-specific LLMs to support personalized treatment recommendations in precision oncology. MEREDITH offers access to research, streamlines data analysis, and generates comprehensive treatment options, including those informed by preclinical data and combination therapies. However, human expertise remains irreplaceable for clinical judgment and treatment selection within MTBs. Future research should focus on the ethical integration of LLMs into clinical workflows. This requires robust data privacy safeguards, XAI to build trust, and transparency in MTB deliberations. By addressing these considerations, we can pave the way for the responsible integration of LLMs, potentially improving patient outcomes while upholding their privacy and autonomy.

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APPENDIX 1. ITERATIVE DEVELOPMENT PROCESS OF THE LARGE LANGUAGE MODEL SYSTEM MEDICAL EVIDENCE RETRIEVAL AND DATA INTEGRATION FOR TAILORED HEALTHCARE

A key factor to improve accuracy and reliability of large language model (LLM)–generated recommendations was significantly reducing the risk of hallucinations by grounding it in context using *retrieval-augmented generation* (*RAG*) techniques, leveraging metadata, and using *chain of thought* (*CoT*) prompting.

This approach yielded several benefits.

Grounding: The incorporation of RAG served to mitigate the potential for artifactual outputs, which were notably absent in the evaluations conducted by MTB experts on both the draft and enhanced LLM recommendations. Expanding the corpus with diverse data sources beyond the specific molecular profile (*PubMed*-indexed data, oncology guidelines, clinical trial databases, and drug availability databases) led to more grounded, factual, and comprehensive treatment recommendations.

Metadata: Metadata such as the type of publication, date of publication, and evidence graduation of the research enabled us to retrieve more relevant information than simple semantic similarity search.

CoT: Chain of thought is a prompting technique that mimics human expert thought processes in sequential steps to achieve better results. Using *CoT* prompting techniques with four consecutive prompts (identify relevant research, identify applicable clinical trials, identify matching oncologic guidelines, and prioritize treatment recommendations) improved response quality and created responses that better reflected human expert reasoning. We intentionally encouraged the model to provide more and diverse treatment recommendations, relying on human experts to review and prioritize suitable options for patients.

Benchmark: Human Expert Recommendations

Session 1: January 10, 2024

Case 1:

Limited treatment options: Because of the current tumor profile, no definitive treatment recommendation can be made at this time. After completion of guideline-aligned therapy, explore potential clinical trial options tailored to the patient's specific treatment history.

Genetic referral: Given the patient's age (56 years) and lack of details on family history, a cautious approach to human genetic referral is recommended.

1. *TP53* mutation without variant allele frequency (VAF) data is considered inconclusive from a human genetic perspective.
2. *CDH1* mutation is a variant of uncertain significance.

Case 2:

Genetic referral: Because of the young age of onset (26 years) for urachal carcinoma, human genetic testing is recommended despite a lack of reported family history.

1. *TP53* mutation: This mutation cannot be assessed from a human genetic perspective without VAF data.

Treatment options:

1. After completion of guideline-directed therapy, or even during treatment, evaluate potential immunotherapy trials on the basis of the patient's previous therapies and response to immunotherapy.
2. Considering the *PIK3CA* amplification, treatment with everolimus could be a potential option.

Case 3:

Genetic referral: *ATM* variant (class 4, incomplete nucleotide information) suggests consideration of human genetic consultation. *BRCA2* variant is classified as benign.

Treatment options:

1. Human epidermal growth factor receptor 2 (HER2) inhibition: Consider trastuzumab deruxtecan (evidence level at least m2A, DESTINY-04). An additional HER2 immunohistochemistry (IHC) test is recommended. Conduct a literature search for relevant case reports.
2. *ATM*: If HRD testing reveals a positive result and the patient has not progressed on platinum-based chemotherapy, consider poly (ADP-ribose) polymerase inhibitor therapy.
3. *cMET* inhibition: Evaluate *cMET* expression via IHC. If positive, consider telisotuzumab vedotin (or capmatinib as an alternative).
4. *FGFR3* overexpression: Consider futibatinib for *FGFR3* overexpression (evidence level m2B, PMID: 34551969).
5. Clinical trial options: After completing guideline-directed therapy, explore potential clinical trials tailored to the patient's previous therapies and response.

Case 4:

Treatment options:

1. *PIK3CA* mutation: Consider enrollment in the CRAFT trial (ClinicalTrials.gov identifier: NCT04551521) after chemotherapy failure. Alternatively, PIK3CA inhibitor alpelisib can be explored (PMIDs: 29401002, 31091374, and 35133871).
2. *FGFR3* overexpression: If the *FGFR3* variant is confirmed pathogenic, futibatinib may be an option after completion of guideline-directed therapy (PMID: 34551969).
3. Clinical Trials: After completion of guideline-directed therapy, explore potential clinical trial options tailored to the patient's specific treatment history.

Case 5:

Treatment options:

1. Clinical Trials: Explore enrollment in clinical trials (eg, ClinicalTrials.gov identifiers: NCT02864992, NCT03539536, NCT03940703).
2. *MET* Amplification: MET amplification is a potential secondary resistance mechanism in *EGFR*-inhibitor treated patients.
3. If no *EGFR* inhibitors have been used, consider *EGFR* inhibition on the basis of the level of *MET* amplification and clinical progression. However, if resistance is mediated by *MET*, use *EGFR* inhibitors with caution (PMID: 34837746).
4. *cMET*-targeted therapy: Depending on the specific situation, consider telisotuzumab vedotin (early access program Gauting) or capmatinib (evidence level m1A)[32] for *cMET*-targeted therapy.

Session 2: January 17, 2024

Case 6:

Treatment options:

1. *KRAS* p.G12C mutation (VAF 18%): Consider sotorasib, which is approved from the second line of therapy onward (evidence level).[33]
2. Consider enrollment in clinical trials (*nNGM*), such as KontRASt-01 and KontRASt-03 (JGQ; Freiburg, Würzburg).

Case 7:

Genetic referral: Because of the presence of *NF1* and *TP53* mutations, human genetic consultation is recommended.

Treatment options:

1. *NF1* Mutation (p.I1605fs, VAF 39%): Consider MEK inhibition with trametinib after completion of guideline-directed therapy. This recommendation is based on a case report demonstrating partial response in *NF1*-mutated melanoma treated with trametinib (evidence level m1C).[34]

Case 8:

Treatment options:

1. *FGFR2* translocation: Consider futibatinib or pemigatinib, both approved therapies by the EMA.
2. Clinical trial option: If the patient is treatment-naïve, explore enrollment in the REFOCUS clinical trial.

Case 9:

1. *PIK3CA* mutation (p.E545K, VAF 39%): A case report suggests potential benefit from combining alpelisib (PI3K inhibitor) with bicalutamide (androgen receptor antagonist) in *PIK3CA*-mutated tumors with high AR expression (evidence level m1C, PMID: 34036229).
2. The presence of a concomitant *HRAS* mutation may reduce the efficacy of *PIK3CA*-targeted therapy. Additional references: PMID: 29401002, 31091374, and 35133871.
3. Tipifarnib (farnesyltransferase inhibitor) demonstrated promising results in a Phase II trial for head and neck squamous cell carcinoma (overall response rate 55%, median progression-free survival 5.6 months, PMID: 33750196).

Treatment options and additional diagnostics:

1. Perform additional IHC analysis for HER2 and AR to inform treatment decisions.
2. Clinical trials: Consider enrollment in the CRAFT trial (ClinicalTrials.gov identifier: NCT04551521) for *PIK3CA*-targeted treatment after chemotherapy failure, even with the presence of the *HRAS* mutation. Discuss the potential benefits and drawbacks with the patient.
3. Discuss the possibility of alpelisib therapy despite the *HRAS* mutation, acknowledging the potential for reduced efficacy.
4. Consider tipifarnib as an alternative targeted therapy option.
5. Immunotherapy because of high tumor mutation burden: potential benefit of pembrolizumab.
6. Explore potential inclusion in the MASTER program

Case 10:

EGFR mutations:

1. Exon 19 deletion (p.E746_A750del, VAF 50%): This mutation is associated with reduced tumor infiltration by $CD8^+$ T cells.[35]
2. Tyrosine kinase inhibitor (TKI) resistance mutation (p.C797S, VAF 29%): This mutation may confer resistance to third-line *EGFR* TKIs.[36]

STK11 mutation (p.C210, VAF 39%): This pathogenic mutation suggests possible primary resistance to PD-1 blockade therapy (ASCO abstract).[37]

Treatment options:

1. *EGFR* resistance testing: Consider osimertinib resistance testing to inform further treatment decisions.
2. Continue with guideline-directed therapy.
3. Clinical trial options: Explore potential enrollment in relevant clinical trials on the basis of the patient's specific tumor characteristics and *EGFR* resistance profile (48 clinical trials listed on QuickQueck.de).

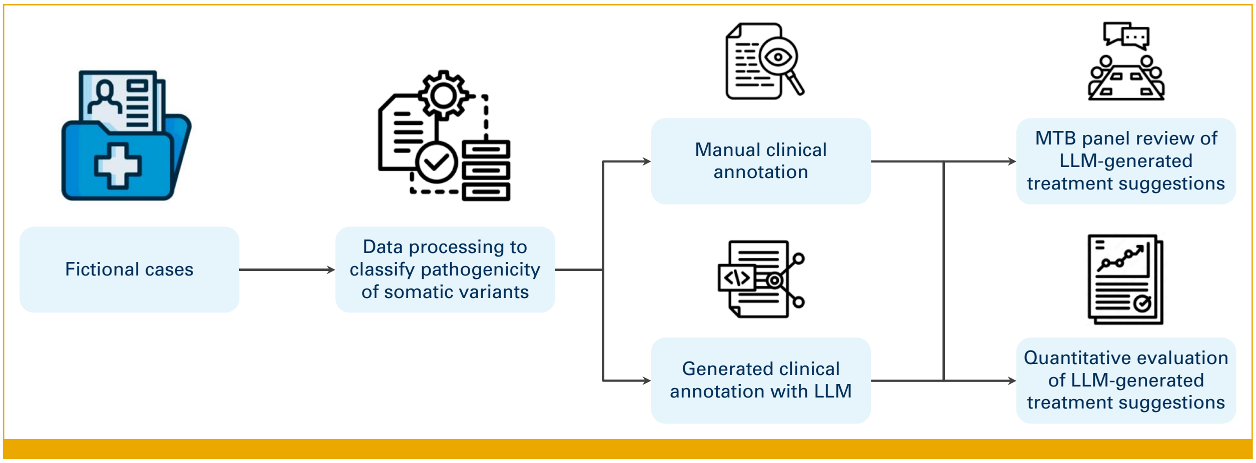

FIG A1. General workflow of the study. Free icons from Flaticon[38] were used. LLM, large language model.

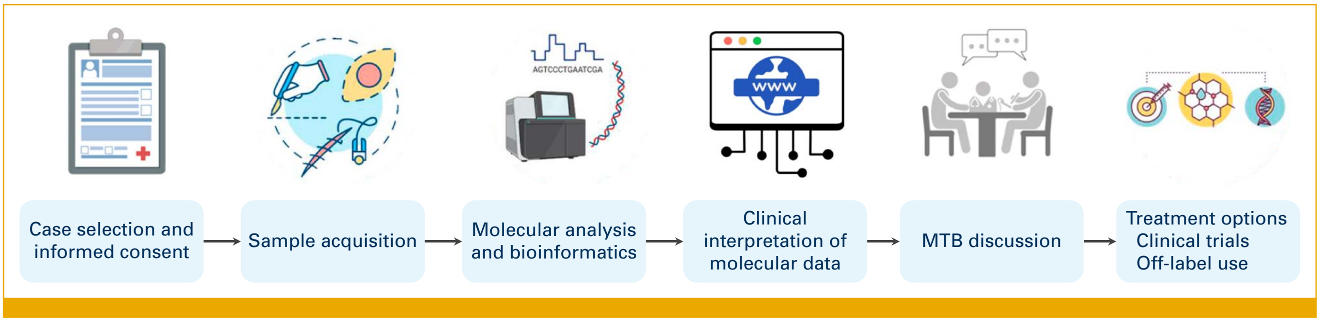

FIG A2. Operational workflow of the MTB at the ZPM^{TUM} facility. Free icons from Flaticon[38] were used. MTB, molecular tumor board.

TABLE A1. Description of Fictional Patients

No.	Age, Years	Sex	Diagnosis	Stage	Sequencing Type	Molecular Alterations ([likely] pathogenic per molecular tumor board panel]
1	56		Lung adenocarcinoma	IV	Panel	***KRAS* p.G13D**, ***TP53* p.A276G**, *PTPRS* R238*, *ZFHX3* p.F2994L, *CDH1* p.D433N
2	26		Urachal carcinoma	IV	Panel	***KRAS* p.G12V**, *BCORL* p.R1332*, ***TP53* p.H214fs*7**, *CDKN2C* p.L65F, *MAP3K1* p.T949_E950insT, ***MYCN* p.E47fs*8**, *CTNNA1* p.K577_L578 > TKL, *JAK1* p.I597M, *FANCL* p.T367fs*12+, ***PIK3CA* amplification (n > 6)**, ***MYC* amplification (n > 6)**, ***MYCL1* amplification (n > 6)**, ***SOX2* amplification (n > 6)**, ***MUTYH* amplification (n > 6)**
3	58		Thymic adenocarcinoma	IV	Whole-exome	Germline: *BRCA2* p.K3326* (1N), tumor: ***SMAD4* p.C363R (1N)**, *TP53* p.305fs (2N_LOH), *CDKN1B* p.K100N (2N_LOH), ***ATM* p.E1666* (4N)**, *MAP3K8* p.H236Y (1N), *TRAF1* p.R70H (2N), *HDAC2* p.R409* (1N), *TMEM111-TDRD3* fusion, ***PRKDC-CDH17* fusion**, ***EXT1-MAGI2* fusion**, ***ERBB2* RNA overexpression (RPKM tumor 45 *v* 1.8 control)**, ***ERBB3* RNA overexpression (RPKM tumor 65.9 *v* 0.2 control)**, ***PDGFRB* RNA overexpression (RPKM tumor 35.8 *v* 2.3 control)**, ***TGFA* RNA overexpression (RPKM tumor 14.2 *v* 0.4 control**, ***EGF* RNA overexpression (RPKM tumor 1.9 *v* 0.1 control)**, ***FGFR3* RNA overexpression (RPKM tumor 11.4 *v* 1.9 control)**, ***MET* RNA overexpression (RPKM tumor 22.1 *v* 1.4 control)**
4	59		Oropharyngeal carcinoma	IV	Panel (tumor purity 60%)	***PIK3CA* p.E545K (AF 25%)**, ***MAPK1* p.E322K (AF 10%)**, *FGFR3* p.D786N (AF 30%)
5	64	w	Lung adenocarcinoma	IV	Panel (tumor purity 30%)	TMB 3.8 Mut/Mb, ***EGFR* p.E746_A750del (AF 43%)**, *TP53* p.A138_Q144del (AF 37%), ***MET* amplification FISH-positive**
6	59	m	Lung adenocarcinoma	IV	Panel (tumor purity 40%)	***KRAS* p.G12C (AF 18%)**, *KEAP1* p.L276F (AF 45%), *STK11* p.K83Tfs*13 (AF 38%)
7	41	w	Melanoma	IV	Panel (tumor purity 80%)	**Tumor mutational burden 12.8 Mut/Mb**, ***NF1* p.I1605fs (AF 39%)**, ***TP53* c.672+1G>A (AF 50%)**, *RB1* p.Q846* (AF 20%), *TERT* p.R859Q (AF 41%)
8	46	m	Cholangiocarcinoma	IV	Panel (tumor purity 80%)	Tumor mutational burden 1.2 Mut/Mb, ***FGFR2::BICC1* fusion**, ***TP53* p.E258* (AF 52%)**
9	79	m	Salivary duct carcinoma	IV	Panel (tumor purity 75%)	**Tumor mutational burden 10.5 Mut/Mb**, ***HRAS* p.Q61R (AF 44%)**, ***PIK3CA* p.E545K (AF 39%)**, *TP53* p.T211A (AF 60%), *KMT2C* p.P2493Q (AF 21%)
10	52	w	Lung adenocarcinoma	IV	Panel (tumor purity 60%)	***EGFR* p.E746_A750del (AF 50%)**, ***EGFR* p.C797S (AF 29%)**, ***STK11* p.C210* (AF 39%)**

NOTE. The entries highlighted in bold are (likely) pathogenic genetic alterations.
Abbreviation: AF, allele frequency.